Part I. SRE in Context

This section provides the foundational background for understanding Site Reliability Engineering (SRE) as it exists today. In the decade since the first edition of this book was published, SRE has evolved from a largely Google-specific methodology into a widespread, mature industry movement. This part of the book sets the stage by defining what modern SRE is, why it represents a strategic business investment, and how our production environments have fundamentally transformed.

In Chapter 1, "Introduction: Modeling What We Do and Why", we break down the very name of the discipline—Site, Reliability, and Engineering—to explain how SREs apply a software engineering mindset to operational problems. We propose a modernized hierarchy of production health—which also serves as organizational framing for this book—that emphasizes psychological safety, continuous improvement, and emergent reliability as the true foundations of healthy production systems.

We then examine the business case for SRE in Chapter 2, "The Value of Reliability". By treating operational management as a software problem rather than a human one, SRE enables sublinear staff scaling and balances feature velocity with system stability. This chapter explores how SRE practices deliver strategic financial value, align product and infrastructure teams, and mitigate the massive direct and opportunity costs of unreliable systems across organizations of all sizes.

Finally, Chapter 3, "Trends in Production", zooms out to examine the broader industry landscape. It explores how cloud computing, the architectural shift from virtual machines to workload-centric containers, and the rise of internal developer platforms have reshaped the baseline expectations for running modern software at scale. This chapter also highlights the transition from measuring reliability via infrastructure metrics to focusing on the actual user experience. It concludes with an overview of the Google pro-

duction environment, to provide context for readers for references used throughout the book.

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